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Electronic Device Having a Supply Terminal and a Thermal Bridge

Abstract

An electronic device includes an enclosure, one or more power semiconductor dies within the enclosure, a supply terminal, and a thermal bridge. The supply terminal is electrically connected to the one or more power semiconductor dies and is at least partially exposed from the enclosure. The supply terminal extends lengthwise in a longitudinal direction and includes a constricted region where a width of the supply terminal reduces in a lateral direction transverse to the longitudinal direction. The thermal bridge is distinct from the enclosure. The thermal bridge bridges the constricted region and provides a thermal conduction path that is in parallel with a thermal conduction path of the supply terminal. The thermal bridge is non-magnetic and has a thermal conductivity that is higher than a thermal conductivity of the enclosure.

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Background/Summary

BACKGROUND

[0001] Demand for electronic components for power applications continues to increase rapidly across a wide range of industries, including automotive, consumer electronics, renewable energy, manufacturing, and medical, among many others. Developments in semiconductor materials such as silicon carbide (SiC) and gallium nitride (GaN) have enabled power electronic components with advantageous features such as smaller footprint, higher voltage and current capabilities, and faster switching speeds. As components become smaller and/or deliver higher currents, efficient heat dissipation becomes particularly important to ensuring acceptable reliability and lifetime. Additionally, many applications of these electronic components benefit from or even require monitoring of current during their operation using a sensor (e.g., a Hall sensor) built into the component. For some electronic component designs, there is a compromise between heat dissipation and the ability to accurately measure current. Thus, there is a need for a solution that enables current to be measured accurately without sacrificing efficient heat dissipation.

SUMMARY

[0002] According to an embodiment of an electronic device, the electronic device comprises: an enclosure; one or more power semiconductor dies within the enclosure; a supply terminal electrically connected to the one or more power semiconductor dies and at least partially exposed from the enclosure, the supply terminal extending lengthwise in a longitudinal direction and comprising a constricted region where a width of the supply terminal reduces in a lateral direction transverse to the longitudinal direction; and a thermal bridge distinct from the enclosure, the thermal bridge bridging the constricted region and providing a thermal conduction path that is in parallel with a thermal conduction path of the supply terminal, wherein the thermal bridge is non-magnetic and has a thermal conductivity that is higher than a thermal conductivity of the enclosure.

[0003] Those skilled in the art will recognize additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0004] The elements of the drawings are not necessarily to scale relative to each other. Like reference numerals designate corresponding similar parts. The features of the various illustrated embodiments can be combined unless they exclude each other. Embodiments are depicted in the drawings and are detailed in the description which follows.

[0005] FIG. **1** illustrates a side cross-sectional view of an electronic device, according to an embodiment.

[0006] FIG. **2** illustrates a top plan view of a supply terminal and a thermal bridge, according to an embodiment.

[0007] FIG. **3** illustrates top plan views of supply terminals and thermal bridges, according to various embodiments.

[0008] FIG. **4** illustrates a side cross-sectional view of an electronic device, according to an embodiment.

[0009] FIG. **5** illustrates a side cross-sectional view of an electronic device, according to an embodiment.

[0010] FIG. **6** illustrates a side cross-sectional view of an electronic device, according to an embodiment.

[0011] FIG. **7** illustrates a side cross-sectional view of an electronic device, according to an embodiment.

[0012] FIG. **8** illustrates a side cross-sectional view of an electronic device, according to an embodiment.

DETAILED DESCRIPTION

[0013] Described herein is an electronic device, such as a molded power semiconductor package or a power semiconductor module, having a supply terminal and associated features that enable the supply terminal to provide both heat dissipation and accurate current measurements during operation of the electronic device. The supply terminal has a constricted region that has a reduced width in a lateral direction relative to the rest of the supply terminal. During operation in which current passes through the supply terminal, higher current density in the constricted region results in a higher magnetic flux produced proximal to the constricted region. The higher magnetic flux near the constricted region of the supply terminal may enable a sensor placed near the constricted region to more accurately measure current through the supply terminal when compared to similar measurements taken elsewhere along the supply terminal where the magnetic flux is lower. However, the narrower constricted region limits the amount of heat energy that can pass through the supply terminal, reducing the efficiency of heat dissipation from the electronic device through the supply terminal.

[0014] According to embodiments described herein, the electronic device includes a thermal bridge that provides a thermal conduction path that is in parallel with the thermal conduction path of the supply terminal. The thermal conduction path through the thermal bridge enables heat energy to bypass the constricted region of the supply terminal. Providing a supply terminal having a constricted region and a thermal bridge that provides a thermal conduction path around the constricted region may, in some examples, enable the supply terminal of the electronic device to both dissipate heat away from the electronic device and provide a magnetic flux that is sufficient to enable accurate current measurements to be taken.

[0015] Described next, with reference to the figures, are exemplary electronic devices having a supply terminal with a constricted region and a thermal bridge that provides a thermal conduction path around the constricted region.

[0016] FIG. **1** illustrates a side cross-sectional view of an electronic device **100**, according to an embodiment. The electronic device **100** includes one or more power semiconductor dies **120** mounted to a substrate **115** within an enclosure **110**.

[0017] In some examples, the enclosure **110** is a molded enclosure that includes a mold compound. In these examples, the electronic device **100** may be a molded power semiconductor package and, although not specifically illustrated in FIG. **1**, the power semiconductor die(s) **120** may be embedded in the enclosure **110**. A mold compound is a plastic encapsulant typically formed from an organic resin such as an epoxy resin. The plastic encapsulant may include fillers such as non-melting inorganic materials. Catalysts may be used to accelerate the cure reaction of the organic resin. Other materials such as flame retardants, adhesion promoters, ion traps, stress relievers, colorants, etc. may be added to the plastic encapsulant, as appropriate. The mold compound may be formed by injection molding, compression molding, film-assisted molding (FAM), reaction injection molding (RIM), resin transfer molding (RTM), blow molding, etc.

[0018] In some examples, the enclosure **110** is a frame enclosure. In these examples, the electronic device **100** may be a power semiconductor module. A frame enclosure may include one or more pieces of metal, plastic, composite, and/or other suitable material that is structured and arranged to enclose the power semiconductor die(s) **120**. A frame enclosure may include a base, a lid, walls, and/or other pieces. A frame enclosure may include openings that provide access to the power semiconductor die(s) **120**, e.g., contact pads. Alternatively, or in addition, a frame enclosure may include feedthrough contacts such as pins or terminals that provide electrical contact to the power semiconductor die(s) **120**.

[0019] The substrate **115** may be a printed circuit board (PCB), lead frame, or other substrate, e.g., insulated metal substrate (IMS), a DCB (direct copper bonded) substrate, an AMB (active metal brazed), etc. The substrate **115** may include one or more layers **117** of insulating material (e.g.

polyimide, ceramic) and one or more metallic layers **119** (e.g., copper, aluminum). The one or more metallic layers **119** may be patterned to provide regions for electrical contact to the power semiconductor die(s) **120**. The substrate **115** may be mounted to a cooler **121** such as a natural or forced convection air cooled heat sink, a single or double sided cooling system with liquid cold plates, etc.

[0020] In FIG. **1**, a bond wire **123** electrically connects the power semiconductor dies **120** (e.g., source contacts of the semiconductor dies **120**) to one another and to one of the metallic layers **119** of the substrate **115**. In other examples, separate bond wires may be used to contact one or more of the power semiconductor die(s) **120** to one another, to one or more metallic layers **119** of the substrate **115**, and/or to another feature of the electronic device **100**. In other examples, other electrical connection means, such as a metallic ribbon or ribbons, clip(s), etc., may be used to connect one or more of the power semiconductor die(s) **120** to one another, to one or more metallic layers **119** of the substrate **115**, and/or to another feature of the electronic device **100**.

[0021] The power semiconductor die(s) **120** may include one or more devices, including transistors, diodes, resistors, capacitors, and/or other types of active or passive devices. Examples of transistors of the power semiconductor die(s) **120** may include MOSFETs, IGBTs, and/or BJTs, among others. The power semiconductor die(s) **120** and/or their constituent devices may be arranged to form all or part of a power electronics circuit such as a DC/AC inverter, a DC/DC converter, an AC/DC converter, a DC/AC converter, an AC/AC converter, a multi-phase inverter, an H-bridge, motor driver, etc. In some examples, a power electronics circuit that includes the power semiconductor die(s) **120** is a half-bridge or full-bridge circuit.

[0022] The electronic device **100** includes a supply terminal **130** electrically connected to one or more of the power semiconductor die(s) **120**, e.g., through one of the metallic layers **119** of the substrate **115**. The supply terminal **130** is at least partially exposed from the enclosure **110**. In the example of the electronic device **100**, the supply terminal **130** extends lengthwise in a longitudinal (current flow) direction x and protrudes from the enclosure **110**. In other examples, a surface of the supply terminal **130** may be exposed from the enclosure **110**. For example, a surface of the supply terminal **130** may be flush with a surface of the enclosure **110**, or may protrude or be recessed from a surface of the enclosure **110**. Other configurations in which the supply terminal **130** is at least partially exposed from the enclosure **110** are contemplated. In examples where the power semiconductor die(s) **120** include one or more transistors (e.g. MOSFET(s), IGBT(s), a half bridge or full bridge circuit), the supply terminal **130** may be a source, emitter, drain, or collector terminal.

[0023] The supply terminal **130** includes a constricted region **132** where a width of the supply terminal **130** reduces in a lateral direction y transverse to the longitudinal direction x. As current passes through the supply terminal **130** along the longitudinal direction x, the current density increases in the constricted region **132** due to the reduced width relative to the rest of the supply terminal **130**. This results in a higher magnetic flux produced proximal to the constricted region **132**. The higher magnetic flux near the constricted region **132** of the supply terminal **130** may enable a sensor placed near the constricted region **132** to more accurately measure current through the supply terminal **130** when compared to similar measurements taken elsewhere along the supply terminal **130** where the magnetic flux is lower.

[0024] FIG. **1** illustrates an exemplary sensor **150** (e.g., a Hall sensor) that is aligned with the constricted region **132** along a z direction that is orthogonal to both the longitudinal direction x and the lateral direction y. In this example, the enclosure **110** includes a recess **112** that is aligned with the constricted region **132** along the z direction and is configured to receive the sensor **150**. The sensor **150** may be attached, e.g., soldered, to a printed circuit board (PCB) **160** (e.g., a gate driver PCB). In other examples, such as a molded enclosure **110**, the sensor **150** may be embedded in the enclosure **110**. The sensor **150** is included herein for illustrative purposes but is optional and is not a requirement of the electronic device **100**.

[0025] According to an embodiment, the electronic device **100** includes a thermal bridge **140**. The thermal bridge **140** is distinct from the enclosure **110** and is electrically isolated from the supply terminal **130**. The thermal bridge **140** thermally bridges the constricted region **132** of the supply terminal **130**, providing a thermal conduction path **172** that is in parallel with a thermal conduction path **171** of the supply terminal **130**. The thermal conduction path **172** through the thermal bridge **140** enables heat energy to bypass the constricted region **132** of the supply terminal **130**, potentially improving heat dissipation from the electronic device **100**. Additionally, by channelling dissipated heat through the thermal bridge **140**, the temperature of the supply terminal **130** near the constricted region **132** may be reduced, thus potentially reducing the impact of the temperature of the constricted region **132** on the sensor **150** and enabling the sensor **150** to provide more accurate current measurements and/or avoiding excessive heating of the sensor **150**.

[0026] The thermal bridge **140** is non-magnetic so that the thermal bridge **140** does not affect the magnetic flux around the constricted region **132**, and thus the measurement of the current through the supply terminal **130**. Additionally, the thermal bridge **140** has a thermal conductivity that is higher than a thermal conductivity of the enclosure **110**. For example, the thermal bridge **140** may have a thermal conductivity of at least 20 W/m-K. The thermal bridge **140** may be electrically conductive or electrically insulative since the thermal bridge **140** does not form part of the main current path of the electronic device. In some examples, the thermal bridge **140** is metallic, e.g., comprising copper, aluminum, and/or one or more other metals or metal alloys. The thermal bridge **140** may be formed of a metallic foil. Other example materials of the thermal bridge **140** include a ceramic such as silicon nitride, thermal grease, or a composite material that includes one or more fillers, among other examples.

[0027] In FIG. **1**, the thermal bridge **140** and the constricted region **132** of the supply terminal **130** are embedded in the enclosure **110**. A gap **114** separates the thermal bridge **140** from the supply terminal **130** in the z direction. A portion of the enclosure **110** is disposed in the gap **114**, although in some examples a thermal interface material (e.g., a thermal transfer sheet, thermal filler, foil, glue) may additionally or alternatively be disposed in the gap **114**. The recess **112** that is configured to receive the sensor **150** is on an opposite side of the supply terminal **130** from the thermal bridge **140**. A height h.sub.1 of a gap **116** between a lower surface **112**.sub.S of the recess **112** and a first main surface **130**.sub.S, of the supply terminal **130** is greater than a height h.sub.2 of the gap **114** between a second, opposite main surface **130**.sub.S2 of the supply terminal **130** and a surface **140**.sub.S of the thermal bridge **140** that faces the supply terminal **130**.

[0028] FIG. **2** illustrates a top plan view of a supply terminal **230** and a thermal bridge **240**, according to an embodiment. The supply terminal **230**, a constricted region **232** of the supply terminal **230**, and the thermal bridge **240** are one example of the supply terminal **130**, the constricted region **132**, and the thermal bridge **140**, respectively, of the electronic device **100** of FIG. **1**.

[0029] A width W.sub.ST of the supply terminal **230** reduces to a width W.sub.CR in the lateral direction y in the constricted region **232**. The thermal bridge **240** bridges the constricted region **232** and includes an opening **242** that is aligned with the constricted region **232** such that the constricted region **232** is uncovered by the thermal bridge **240** in the z direction. Specifically, the thermal bridge **240** includes a first segment **244** that bridges the constricted region **232** along a first side of the constricted region **232** in the longitudinal (current flow) x direction and a second segment **246** that is separated from the first segment **244** in the lateral y direction and bridges the constricted region **232** along a second, opposite side of the constricted region **232** in the longitudinal x direction. The opening **242** is between the first segment **244** and the second segment **246** in the region of the constricted region **232**.

[0030] Providing the opening **242** in the thermal bridge **240** and aligned with the constricted region **232** may, in some instances, reduce the presence of stray magnetic fields from the thermal bridge **240** (e.g., from Eddy currents generated from proximity of the thermal bridge **240** to the magnetic field of the supply terminal **230**) in the vicinity of the constricted region **232**. This arrangement may ensure that the magnetic field sensed by a sensor (e.g., the sensor **150** of FIG. **1**) in proximity to the constricted region **232** arises predominately from the supply terminal **230** with little or no interference from the thermal bridge **240**, enabling the sensor to provide a more accurate measurement of current in the supply terminal **230**. Additionally, as the temperature of the thermal bridge **240** increases when dissipating heat from an associated electronic device (e.g., electronic device **100** of FIG. **1**), providing the opening **242** may reduce the impact of the temperature of the thermal bridge **240** on the sensor.

[0031] FIG. **3** illustrates top plan views of supply terminals **330***a-h* and thermal bridges **340***a-h*, according to various embodiments. The supply terminals **330***a-h*, the constricted regions **332***a-h*, and the thermal bridges **340***a-h* of FIGS. **3***a*-**3***h*, respectively, are various examples of the supply terminal **130**, the constricted region **132**, and the thermal bridge **140**, respectively, of the electronic device **100** of FIG. **1**, and of the supply terminal **230**, the constricted region **232**, and the thermal bridge **240**, respectively, of FIG. **2**.

[0032] The examples of FIG. **3** illustrate various orientations of a constricted region of a respective supply terminal. A width of each of the supply terminals **330***a-h* reduces in the lateral direction y in the respective constricted region **332***a-h*. In the examples of the supply terminals **330***a*, **330***c*, **330***d*, **330***e*, and **330***h*, the respective constricted regions **332***a*, **332***c*, **332***d*, **332***e*, and **332***h* are oriented along the longitudinal x direction. In the examples of the supply terminals **330***b*, **330***f*, and **330***g*, the respective constricted regions **332***b*, **332***f*, and **332***g* each include a bend toward the lateral y direction.

[0033] Each thermal bridge **340***a-h* bridges a respective constricted region **332***a-h* and includes a respective opening **342***a-h* that is aligned with the respective constricted region **332***a-h* such that the respective constricted region **332***a-h* is uncovered by the respective thermal bridge **340***a-h* in the z direction. Each thermal bridge **340***a-h* includes a respective first segment **344***a-h* that bridges the respective constricted region **332***a-h* along a first side of the respective constricted region **332***a-h* in the longitudinal (current flow) x direction and a respective second segment **346***a-h* that is separated from the respective first segment **344***a-h* in the lateral y direction and bridges the respective constricted region **332***a-h* along a second, opposite side of the respective constricted region **332***a-h* in the longitudinal x direction. The respective opening **342***a-h* is between the respective first segment **344***a-h* and the respective second segment **346***a-h*. In the examples of the thermal bridges **340***a*, **340***b*, and **340***c*, the first segment **344***a*, **344***b*, and **344***c*, respectively, and the second segment **346***a*, **346***b*, and **346***c*, respectively, of the thermal bridge **340***a*, **340***b*, and **340***c*, respectively, are separate pieces.

[0034] FIGS. **3***a*-**3***h* illustrate only a few example configurations of the supply terminal, the constricted region, and the thermal bridge, according to embodiments of the present disclosure. These examples are not limiting, and other structures, arrangements, orientations, features, etc. of supply terminals, constricted regions, and thermal bridges are contemplated and are within the scope of the present disclosure.

[0035] FIG. **4** illustrates a side cross-sectional view of an electronic device **400**, according to an embodiment. Unless specifically noted herein, the electronic device **400** may be similar to the electronic device **100** of FIG. **1**. Similarities to the electronic device **100** may include structure, composition, function, features, components, dimensions, and/or other attributes of the electronic device **100** and its features.

[0036] The electronic device **400** includes an enclosure **410**, a supply terminal **430** having a constricted region **432**, and a thermal bridge **440** that bridges the constricted region **432** of the supply terminal **430**. The enclosure **410** may be similar to the enclosure **110** of FIG. **1**. The supply terminal **430** may be electrically connected to one or more power semiconductor dies (e.g., the power semiconductor die(s) **120** of FIG. **1**). The supply terminal **430** and the constricted region **432** may be similar to the supply terminal **130** and the constricted region **132** of FIG. **1**, the supply terminal **230** and the constricted region **232** of FIG. **2**, one of the supply terminals **330***a-h* and the constricted regions **332***a-h* of FIG. **3**, or may be different than these. The thermal bridge **440** may be similar to the thermal bridge **140** of FIG. **1**, the thermal bridge **240** of FIG. **2**, one of the thermal bridges **340***a-h* of FIG. **3**, or may be different than these. FIG. **4** illustrates an exemplary sensor **450** that may be similar to the sensor **150** of FIG. **1**. The sensor **450** is attached to an exemplary PCB **460** and is aligned with the constricted region **432** along the z direction. The enclosure **410** includes a recess **412** that is configured to receive a sensor, e.g., the sensor **450**. The sensor **450**, the PCB **460**, and the recess **412** are included herein for illustrative purposes but are optional and are not requirements of the electronic device **400**.

[0037] The electronic device **400** differs from the electronic device **100** of FIG. **1** in that a part of the thermal bridge **440** is attached to the supply terminal **430**. Specifically, the supply terminal **430** includes a first portion **430**, and a second portion **430**.sub.2 on opposite sides of the constricted region **432**. A thickness t.sub.2 of the second portion **430**.sub.2 of the supply terminal **430** is greater than a thickness t.sub.1 of the first portion **430**, of the supply terminal **430** such that a gap **414** separates a first portion **440**, of the thermal bridge **440** from the first portion **430**, and the constricted region **432** of the supply terminal **430**. A second portion **440**.sub.2 of the thermal bridge **440** is attached to the second portion **430**.sub.2 of the supply terminal **430**. The second portion **440**.sub.2 of the thermal bridge **440** may be attached to the second portion **430**.sub.2 of the supply terminal **430** by a thermally conductive material such as an adhesive or glue, or may be soldered, diffusion soldered, welded, brazed, or clinched to the second portion **430**.sub.2.

[0038] In FIG. **4**, a material **480** is disposed in the gap **414**. The material **480** may be a part of the enclosure **410** (e.g., a mold compound or part of a frame of the enclosure **410**), or may be a thermal interface material that is distinct from the enclosure **410**, such as a thermal transfer sheet, thermal filler, foil, glue, or other suitable material.

[0039] FIG. **5** illustrates a side cross-sectional view of an electronic device **500**, according to an embodiment. Unless specifically noted herein, the electronic device **500** may be similar to the electronic device **100** of FIG. **1**. Similarities to the electronic device **100** may include structure, composition, function, features, components, dimensions, and/or other attributes of the electronic device **100** and its features.

[0040] The electronic device **500** includes an enclosure **510**, a supply terminal **530** having a constricted region **532**, and a thermal bridge **540** that bridges the constricted region **532** of the supply terminal **530**. The enclosure **510** may be similar to the enclosure **110** of FIG. **1**. The supply terminal **530** may be electrically connected to one or more power semiconductor dies (e.g., the power semiconductor die(s) **120** of FIG. **1**). The supply terminal **530** and the constricted region **532** may be similar to the supply terminal **130** and the constricted region **132** of FIG. **1**, the supply terminal **230** and the constricted region **232** of FIG. **2**, one of the supply terminals **330***a-h* and the constricted regions **332***a-h* of FIG. **3**, or may be different than these. The thermal bridge **540** may be similar to the thermal bridge **140** of FIG. **1**, the thermal bridge **240** of FIG. **2**, one of the thermal bridges **340***a-h* of FIG. **3**, or may be different than these. FIG. **5** illustrates an exemplary sensor **550** that may be similar to the sensor **150** of FIG. **1**. The sensor **550** is attached to an exemplary PCB **560** and is aligned with the constricted region **532** along the z direction. The enclosure **510** includes a recess **512** that is configured to receive a sensor, e.g., the sensor **550**. The sensor **550**, the PCB **560**, and the recess **512** are included herein for illustrative purposes but are optional and are not requirements of the electronic device **500**.

[0041] The electronic device **500** differs from the electronic device **100** of FIG. **1** in that a part of the thermal bridge **540** is attached to the supply terminal **530**. Specifically, the supply terminal **530** includes a first portion **530**, and a second portion **530**.sub.2 on opposite sides of the constricted region **532**. The supply terminal **530** has a bend between the first portion **530**, and the second portion **530**.sub.2 such that a gap **514** separates a first portion **540**, of the thermal bridge **540** from the first portion **530**, and the constricted region **532** of the supply terminal **530**. A second portion **540**.sub.2 of the thermal bridge **540** is attached to the second portion **530**.sub.2 of the supply terminal **530**. The second portion **540**.sub.2 of the thermal bridge **540** may be attached to the second portion **530**.sub.2 of the supply terminal **530** by a thermally conductive material such as an adhesive or glue, or may be soldered, diffusion soldered, welded, brazed, or clinched to the second portion **530**.sub.2.

[0042] In FIG. **5**, a material **580** is disposed in the gap **514**. The material **580** may be a part of the enclosure **510** (e.g., a mold compound or part of a frame of the enclosure **510**), or may be a thermal interface material that is distinct from the enclosure **510**, such as a thermal transfer sheet, thermal filler, foil, glue, or other suitable material.

[0043] FIG. **6** illustrates a side cross-sectional view of an electronic device **600**, according to an embodiment. Unless specifically noted herein, the electronic device **600** may be similar to the electronic device **100** of FIG. **1**. Similarities to electronic device **100** may include structure, composition, function, features, components, dimensions, and/or other attributes of the electronic device **100** and its features.

[0044] The electronic device **600** includes an enclosure **610**, a supply terminal **630** having a constricted region **632**, and a first thermal bridge **640** on a first side of the supply terminal **630**. The first thermal bridge **640** bridges the constricted region **632** of the supply terminal **630** and provides a first thermal conduction path **672** that is in parallel with a thermal conduction path **671** of the supply terminal **630**. The enclosure **610** may be similar to the enclosure **110** of FIG. **1**. The supply terminal **630** may be electrically connected to one or more power semiconductor dies (e.g., power semiconductor die(s) **120** of FIG. **1**). The supply terminal **630** and the constricted region **632** may be similar to the supply terminal **130** and the constricted region **132** of FIG. **1**, the supply terminal **230** and the constricted region **232** of FIG. **2**, one of the supply terminals **330***a-h* and the constricted regions **332***a-h* of FIG. **3**, or may be different than these. The first thermal bridge **640** may be similar to the thermal bridge **140** of FIG. **1**, the thermal bridge **240** of FIG. **2**, one of the thermal bridges **340***a-h* of FIG. **3**, or may be different than these. FIG. **6** illustrates an exemplary sensor **650** that may be similar to the sensor **150** of FIG. **1**. The sensor **650** is attached to an exemplary PCB **660** is aligned with the constricted region **632** along the z direction. The enclosure **610** includes a recess **612** that is configured to receive a sensor, e.g., the sensor **650**. The sensor **650**, the PCB **660**, and the recess **612** are included herein for illustrative purposes but are optional and are not requirements of the electronic device **600**.

[0045] The electronic device **600** differs from the electronic device **100** in that it includes a second thermal bridge **641** that is distinct from the enclosure **610** and is on a second, opposite side of the supply terminal **632** from the first thermal bridge **640**. The second thermal bridge **641** bridges the constricted region **632** and provides a second thermal conduction path **673** that is in parallel with the thermal conduction path **671** of the supply terminal **630**. A first gap **614** separates the first thermal bridge **640** from the supply terminal **630** and a second gap **616** separates the second thermal bridge **641** from the supply terminal **630**. In FIG. **6**, the first gap **614** and the second gap **616** each include a portion of the enclosure **610**. In other examples, one or both of the first gap **614** and/or the second gap **616** may include a thermal interface material that is distinct from the enclosure **610** (e.g., a thermal transfer sheet, thermal filler, foil, glue, etc.).

[0046] The second thermal bridge **641** includes an opening **643** that is aligned with the constricted region **632** along the z direction. The opening **643** of the second thermal bridge **641** may be aligned with an opening of the first thermal bridge **640** along the z direction.

[0047] The second thermal bridge **641** is non-magnetic and has a thermal conductivity that is higher than a thermal conductivity of the enclosure **610**. The structure, composition, features, and dimensions, e.g., shape and dimensions of the opening **643**, and/or other attributes of the second thermal bridge **641** may be similar to those of other thermal bridges described herein, e.g., the thermal bridge **140** of FIG. **1**, the thermal bridge **240** of FIG. **2**, one of the thermal bridges **340***a-h* of FIG. **3**, the first thermal bridge **640** of FIG. **6**, or may be different than these.

[0048] While the first thermal bridge **640** and the second thermal bridge **641** of this example are separated from the supply terminal **630**, examples in which at least part of the first thermal bridge **640** and/or the second thermal bridge **641** is attached to the supply terminal **630** are contemplated, similar to the examples of the thermal bridges **440** and **540** illustrated in FIGS. **4** and **5**, respectively.

[0049] FIG. **7** illustrates a side cross-sectional view of an electronic device **700**, according to an embodiment. FIG. **7** illustrates an alternative arrangement of the electronic device **600** of FIG. **6**. The features illustrated in FIG. **7** are similar to the correspondingly numbered features illustrated in FIG. **6**. The electronic device **700** of FIG. **7** differs from the electronic device **600** of FIG. **6** in that the recess **612** extends at least partly through the opening **643** such that the second thermal bridge **641** surrounds the recess **612**. Additionally, thermal interface materials **780** and **781** may be disposed in the first gap **614** and/or the second gap **616**, respectively. Each of the thermal interface materials **780** and **781** may be a thermal transfer sheet, thermal filler, foil, glue, or other suitable material. In other examples, one or both of the first gap **614** and/or the second gap **616** may be partially or completely filled with a portion of an enclosure (e.g., the enclosure **610** of FIG. **6**). In some examples, first thermal bridge **640**, second thermal bridge **641**, the constricted region **632** of the supply terminal **630**, and the thermal interface materials **780** and **781** are partially or completely outside of an enclosure of the electronic device **700** (e.g., the enclosure **610** of FIG. **6**).

[0050] FIG. **8** illustrates a side cross-sectional view of an electronic device **800**, according to an embodiment. Unless specifically noted herein, the electronic device **800** may be similar to the electronic device **100** of FIG. **1**. Similarities to electronic device **100** may include structure, composition, function, features, components, dimensions, and/or other attributes of the electronic device **100** and its features.

[0051] The electronic device **800** includes one or more power semiconductor dies **820** attached to a substrate **815** within an enclosure **810**, a supply terminal **830** having a constricted region **832**, and a thermal bridge **840**. The supply terminal **830** is electrically connected to one or more of the power semiconductor die(s) **820**. The first thermal bridge **840** bridges the constricted region **832** of the supply terminal **830**.

[0052] The enclosure **810** may be similar to the enclosure **110** of FIG. **1**. The substrate **815** may be similar to the substrate **115** of FIG. **1**. The substrate **815** may include one or more layers **817** of insulating material (e.g. polyimide, ceramic) and one or more metallic layers **819** (e.g., copper, aluminum). The layers **817** and **819** may be similar to the layers **117** and **119**, respectively, of FIG. **1**. The substrate **815** may be mounted to a cooler **821**. The cooler **821** may be similar to the cooler **121** of FIG. **1**. A bond wire **823** electrically connects the power semiconductor dies **820** (e.g., source contacts of the semiconductor dies **820**) to one another and to one of the metallic layers **819** of the substrate **815**. As in the example of electronic device **100** of FIG. **1**, other examples may include separate bond wires to contact one or more of the power semiconductor die(s) **820** to one another, to one or more metallic layers **819** of the substrate **815**, and/or to another feature of the electronic device **800**, and/or may include other electrical connection means, such as a metallic ribbon or ribbons, clip(s), etc., to connect one or more of the power semiconductor die(s) **820** to one another, to one or more metallic layers **819** of the substrate **815**, and/or to another feature of the electronic device **800**.

[0053] The power semiconductor die(s) **820** may be similar to the power semiconductor die(s) **120** of FIG. **1**. The supply terminal **830** and the constricted region **832** may be similar to the supply terminal **130** and the constricted region **132** of FIG. **1**, the supply terminal **230** and the constricted region **232** of FIG. **2**, one of the supply terminals **330***a-h* and the constricted regions **332***a-h* of FIG. **3**, or may be different than these. The thermal bridge **840** may be similar to the thermal bridge **140** of FIG. **1**, the thermal bridge **240** of FIG. **2**, one of the thermal bridges **340***a-h* of FIG. **3**, or may be different than these. While the thermal bridge **840** of this example is separated from the supply terminal **830**, examples in which at least part of the thermal bridge **840** is attached to the supply terminal **830** are contemplated, similar to the examples of the thermal bridges **440** and **540** illustrated in FIGS. **4** and **5**, respectively. Additionally, examples in which a thermal interface material (e.g., a thermal transfer sheet, thermal filler, foil, glue) is disposed in a gap between the thermal bridge **840** and the supply terminal **830** are contemplated, e.g., as shown in FIGS. **4**, **5** and **7**.

[0054] FIG. **8** illustrates an exemplary sensor **850** that may be similar to the sensor **150** of FIG. **1**. The sensor **850** is aligned with the constricted region **832** along the z direction. The sensor **850** may be electrically coupled to a printed circuit board (PCB) **860** (e.g., a gate driver PCB). The sensor **850** is included herein for illustrative purposes but is optional and is not a requirement of the electronic device **800**.

[0055] The electronic device **800** differs from the electronic device **100** in that the enclosure **810** encloses the one or more power semiconductor dies **820** and a separate electrically insulative body **811** encloses the thermal bridge **840**, the constricted region **832** of the supply terminal **830**, and the sensor **850** (if present). In some examples, the separate electrically insulative body **811** is a molded enclosure that includes a mold compound.

[0056] Although the present disclosure is not so limited, the following numbered examples demonstrate one or more aspects of the disclosure.

[0057] Example 1. An electronic device, comprising: an enclosure; one or more power semiconductor dies within the enclosure; a supply terminal electrically connected to the one or more power semiconductor dies and at least partially exposed from the enclosure, the supply terminal extending lengthwise in a longitudinal direction and comprising a constricted region where a width of the supply terminal reduces in a lateral direction transverse to the longitudinal direction; and a thermal bridge distinct from the enclosure, the thermal bridge bridging the constricted region and providing a thermal conduction path that is in parallel with a thermal conduction path of the supply terminal, wherein the thermal bridge is non-magnetic and has a thermal conductivity that is higher than a thermal conductivity of the enclosure.

[0058] Example 2: The electronic device of example 1, wherein the thermal bridge comprises an opening that is aligned with the constricted region such that the constricted region is uncovered by the thermal bridge in a direction that is orthogonal to both the longitudinal direction and the lateral direction.

[0059] Example 3: The electronic device of example 1 or 2, wherein the thermal bridge comprises a first segment that bridges the constricted region along a first side of the constricted region in the longitudinal direction and a second segment that is separated from the first segment in the lateral direction and bridges the constricted region along a second, opposite side of the constricted region in the longitudinal direction.

[0060] Example 4: The electronic device of any of examples 1 through 3, wherein the first segment and the second segment of the thermal bridge are separate pieces.

[0061] Example 5. The electronic device of any of examples 1 through 4, wherein the thermal bridge is embedded in the enclosure.

[0062] Example 6. The electronic device of any of examples 1 through 5, wherein the thermal bridge is metallic.

[0063] Example 7. The electronic device of any of examples 1 through 6, wherein the thermal bridge is electrically isolated from the supply terminal.

[0064] Example 8. The electronic device of any of examples 1 through 7, wherein a gap separates at least a portion of the thermal bridge from the supply terminal in a direction that is orthogonal to both the longitudinal direction and the lateral direction.

[0065] Example 9. The electronic device of any of examples 1 through 8, wherein a portion of the enclosure is disposed in the gap.

[0066] Example 10. The electronic device of any of examples 1 through 9, wherein a thermal interface material distinct from the enclosure is disposed in the gap.

[0067] Example 11. The electronic device of any of examples 1 through 6 or 8 through 10, wherein the supply terminal comprises a first portion and a second portion on opposite sides of the constricted region, wherein the gap separates a first portion of the thermal bridge from the first portion of the supply terminal, and wherein a second portion of the thermal bridge is attached to the second portion of the supply terminal.

[0068] Example 12: The electronic device of example 11, wherein a thickness of the second portion of the supply terminal is greater than a thickness of the first portion of the supply terminal.

[0069] Example 13. The electronic device of any of examples 1 through 6 or 8 through 12, wherein at least a part of the thermal bridge is attached to the supply terminal.

[0070] Example 14. The electronic device of any of examples 1 through 13, wherein the enclosure comprises a recess that is configured to receive a sensor, and wherein the recess is on an opposite side of the supply terminal from the thermal bridge and is aligned with the constricted region along a direction that is orthogonal to both the longitudinal direction and the lateral direction.

[0071] Example 15. The electronic device of example 14, wherein a height of a first gap between a lower surface of the recess and a first main surface of the supply terminal is greater than a height of a second gap between a second, opposite main surface of the supply terminal and a surface of the thermal bridge that faces the supply terminal.

[0072] Example 16. The electronic device of any of examples 1 through 15, wherein the thermal bridge is a first thermal bridge on a first side of the supply terminal, wherein the thermal conduction path provided by the first thermal bridge is a first thermal conduction path, and wherein the electronic device further comprises a second thermal bridge on a second, opposite side of the supply terminal from the first thermal bridge, wherein the second thermal bridge is distinct from the enclosure, the second thermal bridge bridging the constricted region and providing a second thermal conduction path that is in parallel with the thermal conduction path of the supply terminal, and wherein the second thermal bridge is non-magnetic and has a thermal conductivity that is higher than a thermal conductivity of the enclosure.

[0073] Example 17. The electronic device of any of examples 1 through 16, wherein a first gap separates at least a portion of the first thermal bridge from the supply terminal, and wherein a second gap separates at least a portion of the second thermal bridge from the supply terminal.

[0074] Example 18. The electronic device of any of examples 1 through 17, wherein a thermal interface material distinct from the enclosure is disposed in at least one of the first gap or the second gap.

[0075] Example 19. The electronic device of any of examples 1 through 18, wherein the second thermal bridge comprises an opening that is aligned with the constricted region along a direction that is orthogonal to both the longitudinal direction and the lateral direction, and wherein the enclosure comprises a recess that is configured to receive a sensor, the recess extending at least partly through the opening such that the second thermal bridge surrounds the recess.

[0076] Example 20. The electronic device of any of examples 1 through 19, further comprising a sensor that is that is aligned with the constricted region along a direction that is orthogonal to both the longitudinal direction and the lateral direction.

[0077] Example 21. The electronic device of any of examples 1 through 20, wherein the thermal bridge, the constricted region of the supply terminal, and the sensor are embedded in the enclosure.

[0078] Example 22. The electronic device of any of examples 1 through 4 or 6 through 20, wherein the enclosure encloses the one or more power semiconductor dies, and wherein the electronic device further comprises a separate electrically insulative body enclosing the thermal bridge, the constricted region of the supply terminal, and the sensor.

[0079] Terms such as “first”, “second”, and the like, are used to describe various elements, regions, sections, etc. and are also not intended to be limiting. Like terms refer to like elements throughout the description.

[0080] As used herein, the terms “having”, “containing”, “including”, “comprising” and the like are open ended terms that indicate the presence of stated elements or features, but do not preclude

additional elements or features. The articles “a”, “an” and “the” are intended to include the plural as well as the singular, unless the context clearly indicates otherwise.
[0081] The expression “and/or” should be interpreted to include all possible conjunctive and disjunctive combinations, unless expressly noted otherwise. For example, the expression “A and/or B” should be interpreted to mean only A, only B, or both A and B. The expression “at least one of” should be interpreted in the same manner as “and/or”, unless expressly noted otherwise. For example, the expression “at least one of A and B” should be interpreted to mean only A, only B, or both A and B.
[0082] It is to be understood that the features of the various embodiments described herein can be combined with each other, unless specifically noted otherwise.
[0083] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations can be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.

Claims

1. An electronic device, comprising: an enclosure; one or more power semiconductor dies within the enclosure; a supply terminal electrically connected to the one or more power semiconductor dies and at least partially exposed from the enclosure, the supply terminal extending lengthwise in a longitudinal direction and comprising a constricted region where a width of the supply terminal reduces in a lateral direction transverse to the longitudinal direction; and a thermal bridge distinct from the enclosure, the thermal bridge bridging the constricted region and providing a thermal conduction path that is in parallel with a thermal conduction path of the supply terminal, wherein the thermal bridge is non-magnetic and has a thermal conductivity that is higher than a thermal conductivity of the enclosure.
2. The electronic device of claim 1, wherein the thermal bridge comprises an opening that is aligned with the constricted region such that the constricted region is uncovered by the thermal bridge in a direction that is orthogonal to both the longitudinal direction and the lateral direction.
3. The electronic device of claim 1, wherein the thermal bridge comprises a first segment that bridges the constricted region along a first side of the constricted region in the longitudinal direction and a second segment that is separated from the first segment in the lateral direction and bridges the constricted region along a second, opposite side of the constricted region in the longitudinal direction.
4. The electronic device of claim 3, wherein the first segment and the second segment of the thermal bridge are separate pieces.
5. The electronic device of claim 1, wherein the thermal bridge is embedded in the enclosure.
6. The electronic device of claim 1, wherein the thermal bridge is metallic.
7. The electronic device of claim 1, wherein the thermal bridge is electrically isolated from the supply terminal.
8. The electronic device of claim 1, wherein a gap separates at least a portion of the thermal bridge from the supply terminal in a direction that is orthogonal to both the longitudinal direction and the lateral direction.
9. The electronic device of claim 8, wherein a portion of the enclosure is disposed in the gap.
10. The electronic device of claim 8, wherein a thermal interface material distinct from the enclosure is disposed in the gap.
11. The electronic device of claim 8, wherein the supply terminal comprises a first portion and a second portion on opposite sides of the constricted region, wherein the gap separates a first portion

of the thermal bridge from the first portion of the supply terminal, and wherein a second portion of the thermal bridge is attached to the second portion of the supply terminal.

12. The electronic device of claim 11, wherein a thickness of the second portion of the supply terminal is greater than a thickness of the first portion of the supply terminal.

13. The electronic device of claim 1, wherein at least a part of the thermal bridge is attached to the supply terminal.

14. The electronic device of claim 1, wherein the enclosure comprises a recess that is configured to receive a sensor, and wherein the recess is on an opposite side of the supply terminal from the thermal bridge and is aligned with the constricted region along a direction that is orthogonal to both the longitudinal direction and the lateral direction.

15. The electronic device of claim 14, wherein a height of a first gap between a lower surface of the recess and a first main surface of the supply terminal is greater than a height of a second gap between a second, opposite main surface of the supply terminal and a surface of the thermal bridge that faces the supply terminal.

16. The electronic device of claim 1, wherein the thermal bridge is a first thermal bridge on a first side of the supply terminal, wherein the thermal conduction path provided by the first thermal bridge is a first thermal conduction path, and wherein the electronic device further comprises a second thermal bridge on a second, opposite side of the supply terminal from the first thermal bridge, wherein the second thermal bridge is distinct from the enclosure, the second thermal bridge bridging the constricted region and providing a second thermal conduction path that is in parallel with the thermal conduction path of the supply terminal, and wherein the second thermal bridge is non-magnetic and has a thermal conductivity that is higher than a thermal conductivity of the enclosure.

17. The electronic device of claim 16, wherein a first gap separates at least a portion of the first thermal bridge from the supply terminal, and wherein a second gap separates at least a portion of the second thermal bridge from the supply terminal.

18. The electronic device of claim 17, wherein a thermal interface material distinct from the enclosure is disposed in at least one of the first gap or the second gap.

19. The electronic device of claim 16, wherein the second thermal bridge comprises an opening that is aligned with the constricted region along a direction that is orthogonal to both the longitudinal direction and the lateral direction.

20. The electronic device of claim 1, further comprising a sensor that is that is aligned with the constricted region along a direction that is orthogonal to both the longitudinal direction and the lateral direction.

21. The electronic device of claim 20, wherein the thermal bridge, the constricted region of the supply terminal, and the sensor are embedded in the enclosure.

22. The electronic device of claim 20, wherein the enclosure encloses the one or more power semiconductor dies, and wherein the electronic device further comprises a separate electrically insulative body enclosing the thermal bridge, the constricted region of the supply terminal, and the sensor.